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T11 – 05

Assignment No. 7

Regression Analysis

Regression is a statistical method used in Machine Learning to establish a relationship between independent (input) and dependent (output) variables. It helps in predicting continuous outcomes based on input data. The primary goal of regression is to find the best-fitting line or function that minimizes the error between predicted and actual values.

Hypothesis Function in Regression

The hypothesis function represents the mathematical model used to predict the dependent variable yy based on independent variable(s) xx. For simple linear regression, the hypothesis function is:

$$h_\theta(x) = \theta_0 + \theta_1 x$$

Where:

- $h_\theta(x)$ is the predicted value.
- θ_0 (intercept) and θ_1 (slope) are parameters to be learned.
- x is the input feature.

For multiple linear regression, the hypothesis function extends to:

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ldots + \theta_n x_n$$

Cost Function in Regression

The cost function measures how well the hypothesis function fits the data. It calculates the error between predicted and actual values. The most common cost function for regression is the Mean Squared Error (MSE):

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x_i) - y_i)^2$$

Where:

- m is the number of training examples.
- $h_\theta(x_i)$ is the predicted value.
- y_i is the actual value.

The objective of regression is to minimize $J(\theta)$, which can be done using optimization techniques like **Gradient Descent**.

Types of Regression

1. Linear Regression – Models a linear relationship between input and output.
2. Multiple Linear Regression – Extends linear regression to multiple features.
3. Polynomial Regression – Fits a polynomial curve instead of a straight line.
4. Logistic Regression – Used for classification problems (though called regression).
5. Ridge Regression – Adds L2 regularization to linear regression to prevent overfitting.
6. Lasso Regression – Adds L1 regularization, leading to feature selection.
7. Elastic Net Regression – Combines L1 and L2 regularization.
8. Support Vector Regression (SVR) – Uses Support Vector Machines for regression tasks.
9. Decision Tree Regression – Uses decision trees to predict continuous values.
10. Random Forest Regression – Uses multiple decision trees to improve accuracy.

Gradient Descent Algorithm

Gradient Descent is an optimization algorithm used to minimize the cost function by updating the parameters θ iteratively. The update rule for each parameter is:

$$\theta_j := \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$

Where:

- α is the learning rate.
- $\frac{\partial J(\theta)}{\partial \theta_j}$ is the gradient (partial derivative) of the cost function.

The steps of gradient descent:

1. Initialize θ_0 and θ_1 randomly.
2. Compute the cost function $J(\theta)$.
3. Compute gradients and update parameters.
4. Repeat until convergence.

Program:

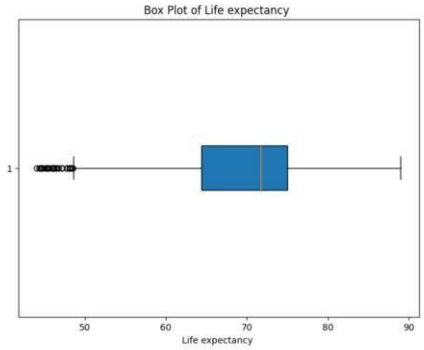
```
import numpy as np import pandas as pd

import matplotlib.pyplot as plt from

sklearn.preprocessing import StandardScaler

from sklearn.preprocessing import

MinMaxScaler data = pd.read_csv("Life

Expectancy Data.csv")


data.shape
```

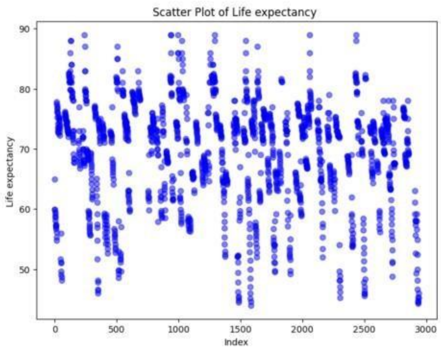
```
data.head()

data.describe()

data.info()
data.isnull().sum()

(data.isnull().sum() / len(data)) * 100

data = data.dropna()

data.isnull().sum()

print(data["Life expectancy "].describe())

column_name = "Life expectancy "

# Create a box plot
plt.figure(figsize=(8, 6))
plt.boxplot(data[column_name].dropna(), vert=False,
patch_artist=True) plt.title(f"Box Plot of {column_name}")
plt.xlabel(column_name) plt.show()

plt.figure(figsize=(8, 6))
```

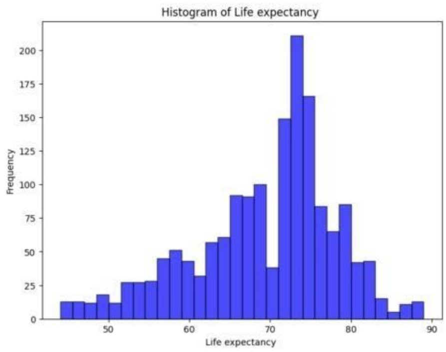
```
plt.scatter(data.index,                 data[column_name],
color='blue',   alpha=0.5)   plt.title(f"Scatter   Plot   of
{column_name}")                              plt.xlabel("Index")
plt.ylabel(column_name) plt.show()


plt.figure(figsize=(8, 6))
plt.hist(data[column_name].dropna(), bins=30, color='blue', edgecolor='black',
alpha=0.7)
plt.title(f"Histogram of
{column_name}")
plt.xlabel(column_name)
plt.ylabel("Frequency") plt.show()


print(data["Life expectancy "].value_counts(normalize=True))


plt.figure(figsize=(8, 6))
plt.hist(data["Life expectancy "].dropna(), bins=30, color='blue',
edgecolor='black',
alpha=0.7)
plt.title("Distribution of Life
Expectancy") plt.xlabel("Life
Expectancy") plt.ylabel("Frequency")
plt.show()
```

Output:

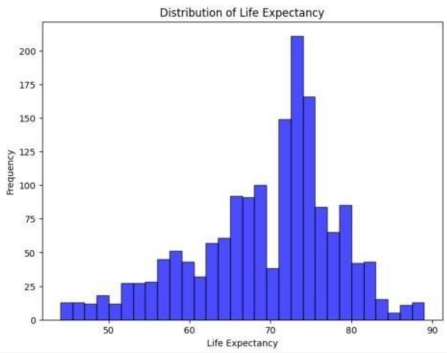
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2938 entries, 0 to 2937
Data columns (total 22 columns):
 #   Column                           Non-Null Count  Dtype  
---  ------                           --------------  -----  
 0   Country                          2938 non-null   object 
 1   Year                             2938 non-null   int64  
 2   Status                           2938 non-null   object 
 3   Life expectancy                  2928 non-null   float64
 4   Adult Mortality                  2928 non-null   float64
 5   infant deaths                    2938 non-null   int64  
 6   Alcohol                          2744 non-null   float64
 7   percentage expenditure           2938 non-null   float64
 8   Hepatitis B                      2385 non-null   float64
 9   Measles                          2938 non-null   int64  
 10   BMI                             2904 non-null   float64
 11  under-five deaths                2938 non-null   int64  
 12  Polio                            2919 non-null   float64
 13  Total expenditure                2712 non-null   float64
 14  Diphtheria                       2919 non-null   float64
 15   HIV/AIDS                        2938 non-null   float64
 16  GDP                              2490 non-null   float64
 17  Population                       2286 non-null   float64
 18   thinness  1-19 years            2904 non-null   float64
 19   thinness 5-9 years              2904 non-null   float64
```








```
Life expectancy
73.0    0.021831
75.0    0.012129
73.9    0.011522
78.0    0.010309
74.5    0.010309
          ...   
55.8    0.000606
58.7    0.000606
49.5    0.000606
82.7    0.000606
44.3    0.000606
Name: proportion, Length: 320, dtype: float64
```



Conclusion:

Regression is a fundamental concept in Machine Learning for predicting continuous values. The hypothesis function models the relationship, while the cost function helps evaluate model performance. Optimization techniques like Gradient Descent improve the accuracy of regression models by minimizing the cost function iteratively. Various types of regression exist, each suited for different types of data and applications.